

Curriculum Vitae ir. Matthias Moulin

Personalia

Nationality: Belgian Birthdate: 15 January 1992
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Experience (in reverse chronological order)

Oct 2025 – Present [Advanced Micro Devices](#) (AMD), Brussels, Belgium – Remote
MTS Rendering Software Development Engineer – GPU and Gaming Team
• **Technologies:** AMD FidelityFX, AMD FSR Ray Regeneration

Dec 2023 – Jun 2025 [Frostbite](#), [EA Digital Illusions CE AB](#) (DICE), Stockholm, Sweden – Remote
Rendering Software Engineer III – Image Quality (Frostbite Rendering)
• **Technologies++:** Core Render Passes, Gbuffer Layouts

Jun 2020 – Nov 2023 [Frostbite](#), [EA Digital Illusions CE AB](#) (DICE), Stockholm, Sweden – Remote
Rendering Software Engineer II – Image Quality (Frostbite Rendering)
• **Technologies++:** DRS, GI Material, Shader Graphs/Expression Shaders

Feb 2019 – May 2020 [Frostbite](#), [EA Digital Illusions CE AB](#) (DICE), Stockholm, Sweden – On-site
Rendering Software Engineer I – Image Quality (Frostbite Rendering)
• **Technologies:** CPU/GPU Probes, Enlighten, Flux, GI Live Preview, PBR Materials

Oct 2016 – Feb 2019 [Department of Computer Science, KU Leuven](#), Leuven, Belgium – Hybrid
Rendering Research Engineer – [Research Foundation – Flanders](#) (FWO)
• **Fellowship:** PhD Fellowship fundamental research

Oct 2015 – Sep 2016 [Department of Computer Science, KU Leuven](#), Leuven, Belgium – Hybrid
Rendering Research Engineer – [Computer Graphics Research Group \(KU Leuven\)](#)

Education (in reverse chronological order)

2013 – 2015 [KU Leuven](#), Leuven, Belgium
Master of Science in Engineering (Computer Science) — **Magna cum laude** (84.46%)
• **Major:** Human Computer Interaction (Computer Graphics)
• **Thesis:** Hybrid Kd-trees for Photon Mapping and Accelerating Ray Tracing (18.5/20)
Supervisor: prof. dr. ir. Philip Dutré

2010 – 2013 [KU Leuven](#), Leuven, Belgium
Bachelor of Science in Engineering — **Magna cum laude** (76.83%)
• **Major:** Computer Science
• **Minors:** Electrical Engineering and Business Management

Publications (in reverse chronological order)

MOULIN M., DUTRÉ P.: [On the use of Local Ray Termination for Efficiently Constructing Qualitative BSPs, BIHs and \(S\)BVHs](#), *The Visual Computer*, Volume 35, Issue 12, pp. 1809–1826, December 2019 (First online: July 2018).

MOULIN M.: [Hybrid Kd-trees for Photon Mapping and Accelerating Ray Tracing](#), *Master's thesis*, Department of Computer Science, KU Leuven, Belgium, June 2015.

MOULIN M., BILLEN N., DUTRÉ P.: [Efficient Visibility Heuristics for Kd-Trees Using the RTSAH](#), In *Eurographics Symposium on Rendering - Experimental Ideas & Implementations* (June 2015), Lehtinen J., Nowrouzezahrai D., (Eds.), The Eurographics Association, pp. 31–39.

Skills *(selected)*

Frameworks	D3D11, D3D12, DXGI, (EA)STL, MPI, OpenCV, OpenGL, OpenMP
Markup languages	HTML/CSS, Markdeep, Markdown, TeX/LaTeX
Modelling languages	OCL, UML
Programming languages	C (89/90, 99, 11/18, 23), C++ (98/03, 11/14, 17, 20, 23), C#, CUDA C/C++, Erlang, Haskell, Java, JavaScript/TypeScript, Maple, Matlab/Octave, MIPS, Prolog, Python 2/3, Racket
Shading languages	GLSL, HLSL
Version control	Git, Mercurial, Perforce, SVN

Game credits and contributions *(in reverse chronological order)*

Developer(s)	Game	Publisher	Release Date
Pearl Abyss	Crimson Desert	Pearl Abyss	Mar 2026
Treyarch , Raven Software	Call of Duty: Black Ops 7	Activision	Nov 2025
EA DICE , Criterion Games , Motive , Ripple Effect Studios	Battlefield REDSEC	Electronic Arts	Oct 2025
EA DICE , Criterion Games , Motive , Ripple Effect Studios	Battlefield 6	Electronic Arts	Oct 2025
EA Vancouver , EA Romania	EA Sports FC 26	Electronic Arts	Sep 2025
Full Circle	Skate	Electronic Arts	Sep 2025
EA Vancouver	NHL 26	Electronic Arts	Sep 2025
EA Tiburon	Madden NFL 26	Electronic Arts	Aug 2025
EA Tiburon	EA Sports College Football 26	Electronic Arts	Jul 2025
BioWare	Dragon Age: The Veilguard	Electronic Arts	Oct 2024
EA Vancouver	NHL 25	Electronic Arts	Oct 2024
EA Vancouver , EA Romania	EA Sports FC 25	Electronic Arts	Sep 2024
EA Tiburon	Madden NFL 25	Electronic Arts	Aug 2024
EA Tiburon	EA Sports College Football 25	Electronic Arts	Jul 2024
EA Vancouver	EA Sports UFC 5	Electronic Arts	Oct 2023
EA Vancouver	NHL 24	Electronic Arts	Oct 2023
EA Vancouver , EA Romania	EA Sports FC 24	Electronic Arts	Sep 2023
EA Tiburon	Madden NFL 24	Electronic Arts	Aug 2023
EA Tiburon	EA Sports PGA Tour	Electronic Arts	Apr 2023
Motive	Dead Space	Electronic Arts	Jan 2023
Criterion Games	Need for Speed Unbound	Electronic Arts	Nov 2022
EA Vancouver	NHL 23	Electronic Arts	Oct 2022
EA Vancouver , EA Romania	FIFA 23	Electronic Arts	Sep 2022
EA Tiburon	Madden NFL 23	Electronic Arts	Aug 2022
EA DICE , Criterion Games , EA Gothenburg , Ripple Effect Studios	Battlefield 2042	Electronic Arts	Nov 2021
EA Vancouver	NHL 22	Electronic Arts	Oct 2021
EA Vancouver , EA Romania	FIFA 22	Electronic Arts	Sep 2021
EA Tiburon	Madden NFL 22	Electronic Arts	Aug 2021
EA Vancouver , EA Romania	FIFA 21	Electronic Arts	Oct 2020
Motive	Star Wars: Squadrons	Electronic Arts	Oct 2020
EA Tiburon	Madden NFL 21	Electronic Arts	Aug 2020
Ghost Games , Criterion Games	Need for Speed Heat	Electronic Arts	Nov 2019
PopCap Games	Plants vs. Zombies: Battle for Neighborville	Electronic Arts	Oct 2019
EA Vancouver , EA Romania	FIFA 20	Electronic Arts	Sep 2019
EA Tiburon	Madden NFL 20	Electronic Arts	Aug 2019
EA DICE	Battlefield V	Electronic Arts	Nov 2018
EA DICE	Star Wars Battlefront II	Electronic Arts	(post-release) Nov 2017 (post-release)